Sodia Harness: Master Logic & Infrastructure Plan

Ecosystem Architecture | B2C Connective Tissue | Version 1.0.0

The Sodia App ("The Harness") is the single B2C interface connecting consumers to B2B physical infrastructure (GymDash/Clubdash) and independent professional talent (CoachDash). This document outlines the core logic stringing these systems together.

1. Global Infrastructure (The "Agnostic Core")

To maintain an unbroken state across all actors, the ecosystem abandons fractured microservices in favor of a unified real-time architecture.

Frontend Framework

React Native (iOS/Android). Prioritizes a mobile-first, tactile UI with heavy use of bottom sheets to preserve operational context during deep flows.

Database & Backend

Operates on a single, optimized Supabase/PostgreSQL schema. Events logged here immediately reflect across all interconnected dashboards.

Real-Time State

WebSocket subscriptions drive ambient updates. E.g., physical gate clearances instantly bump "Inside Now" capacity limits on the facility tablet.

Offline Resilience

The app caches active passes locally. QR generation falls back to a 60-second TOTP code manually enterable by staff. Tracking queues data locally with idempotency keys.

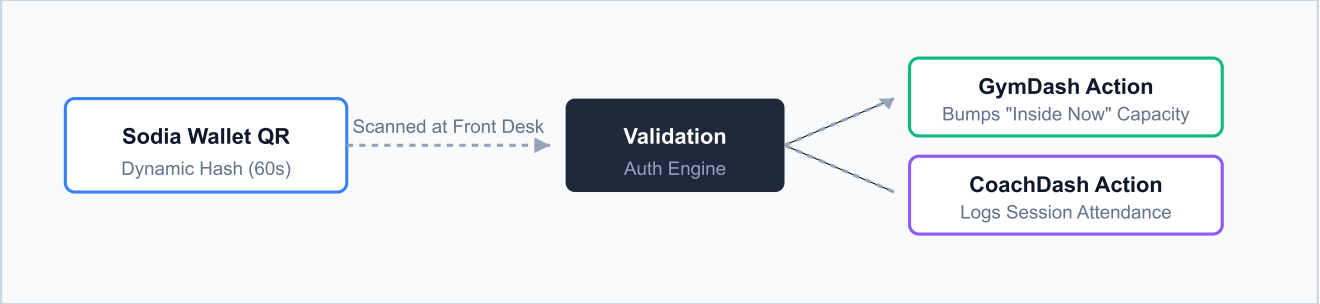
2. Identity & Authentication Logic

Security is tiered based on the friction required for specific actions. A two-token JWT architecture stores refresh tokens in the Secure Enclave.

Friction Level	Method	Governed Actions
Zero Friction	Cached JWT	Launching the app, viewing Sodia feed, generating physical entry QR.
Low Friction	SMS OTP (+63)	Onboarding, deduplicating accounts from Apple/Google SSO fallbacks.

3. Universal Wallet & Gate Logic

The primary interaction between digital identity and physical real estate is the 3D Wallet Stack. Tapping an active pass flips it to reveal a ticking QR code—a dynamic hash of `user_id` + `timestamp` expiring every 60s.



Dual-Authorization: As shown above, if a user has a booked session, one scan clears the facility gate (verifying the active pass) and simultaneously checks the user into the independent trainer's ledger.

4. The Weighted Ranking Engine (WRE) Logic

The Sodia Feed operates as a discovery engine optimized for conversion via deterministic SQL, replacing black-box AI algorithms.

> **Injection Rule:** As users log Trinity Engine data (Move/Lift/Food), the exhaust updates `user_weights`. Every 4th slot in the chronological feed is hijacked by a "Discovery Card" (e.g., Gym Pass, Coach Bounty). Relevancy is multiplied by recent tracker logs.

Actionable Objects: WRE-injected content always renders as a tappable Pill CTA. Tapping never navigates away; it invokes the Universal Checkout bottom sheet directly over the feed.

5. Track Sync Logic (Sodia ↔ CoachDash)

Sodia eliminates the need for separate coaching apps. Coaches build programs via CoachDash, which pushes directly to the `planned_track` table in Sodia.

As the user logs sets, the `actual_track` table populates. The backend compliance algorithm instantly calculates the variance between planned and actual metrics. A 3-day negative delta automatically triggers a check-in message from the coach's profile to the user's inbox, securing retention.

6. Financial Ledger & Split Routing

A hybrid fee model operates over the Xendit xenPlatform API. One-time payments (Day Passes) take a fixed P15 fee, while recurring subscriptions take a flat 5%.

For B2B routing, if a client purchases a \$100 package from an independent coach that mandates a \$10 gym floor fee, the Sodia checkout executes a single charge. The backend ledger instantly fractionates the